

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2025

Course: Bachelor of Technology

Branch : Mechanical

Semester : VI

Subject Code & Name: I.C.Engine BTMEC604B

Max Marks: 60

Date:

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	In Dual cycle, heat addition takes place _____ a) at Constant volume b) first at constant volume then at constant pressure c) constant pressure d. none of the mentioned	CO1	1
2	Choose the correct statement from the following. a) Diesel cycle is more efficient than Otto cycle for a given compression ratio b) Otto cycle is more efficient than Diesel cycle for a given compression ratio c) For a given compression ratio, both Otto and Diesel cycles have same efficiency d. none of the mentioned	CO1	1
3	For same compression ratio and for same heat added _____ a. Otto cycle is more efficient than Diesel Cycle b. Diesel cycle is more efficient than Otto Cycle c. Efficiency depends on other factors d. None of the mentioned	CO1	1
4	The knock in diesel engine occurs due to _____ a. instantaneous and rapid burning of the first part of the charge b. instantaneous auto ignition of last part of charge c. reduction of delay period d. none of the mentioned	CO1	1

5	Ignition quality of petrol is expressed by?				CO1	1
	a) octane number	b) calorific value	c) self ignition temperature	d) cetane number		
6	Octane number is determined by comparing the performance of petrol with the following hydro-carbons?				CO1	1
	a) iso-octane	b) mixture of normal heptane and iso octane	c) alpha methyl naphthalene	d. none of the mentioned		
7	_____ is a colourless gas of about the same density of air.				CO1	1
	a) Carbon monoxide	b)Hydrocarbons	c) Oxides of nitrogen	d) All of the mentioned		
8	_____ is the difference between indicated and brake power of an engine.				CO1	1
	a) Air flow	b) Emissions	c) Friction power	d) None of the mentioned		
9	The pressure of supercharger used is _____				CO1	1
	a)1.0 to 1.3 bar	b) 1.2 to 1.4 bar	c) 1.3 to 1.5 bar	d. none of the mentioned		
10	If the temperature of intake air in internal combustion engine increases, then its efficiency will _____				CO1	1
	a) remain same	b) decrease	c) increase	d) none of the mentioned		
11	The pressure at the end of compression, in petrol engines, is approximately _____				CO1	1
	a) 10 bar	b) 20 bar	c) 25 bar	d) 35 bar		
12	What are the types of Multi-Point Fuel Injection System?				CO1	1
	a) port injection	b) throttle body injection	c) port & throttle body injection	d) none of the mentioned		
Q. 2	Solve the following.					12
A)	Explain with the help of P-θ diagram, different stages of combustion in Spark Ignition engine				CO-3	6
B)	Explain with neat diagram working of four Stroke C. I. engine.				CO-4	6
Q.3	Solve the following.					12
A)	What are the various types of combustion chambers used in SI engines? Explain any one with neat sketch.				CO-4	6
B)	Explain the factors influencing the flame speed with respect to i) Turbulence ii) Fuel- air ratio iii) Compression ratio				CO-3	6

Q. 4	Solve Any Two of the following.		12
A)	State and explain BS-VI norms in detail for CI engines three wheeler and heavy motor vehicle	CO-3	6
B)	Explain exhaust gas recirculation in engine emission control.	CO-4	6
C)	Explain how the supercharging improves the power output of engine? Explain any one type of supercharger with neat sketch	CO-4	6
Q.5	Solve Any Two of the following.		12
A)	Draw the typical performance plot speed characteristics curves for a SI engine and discuss the nature of the curve	CO-3	6
B)	Write a short note on air pollution due to IC engines. List down the sources of pollutants in I.C. Engine? explain its effect on ecology	CO-3	6
C)	Explain hydrogen as alternative fuels for IC engines bringing out their merits and demerits.	CO-4	6
Q. 6	Solve Any Two of the following.		12
A)	Describe the sources for the production of Biodiesel and list down its advantages for utilization in IC engines	CO-3	6
B)	Explain the concept of fuel cell technology.	CO-4	6
C)	Compare Electric vehicle with Hybrid Electric vehicle and list advantages of Hybrid vehicle	CO-4	6
	*** End ***		